REVA UNIVERSITY
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School of Computer Science & Engineering

AI Application Development – IA2

AgroLedger

Team members & roles

TEAM MEMBER NAME	ROLE
NISHA YADAV	TEAM LEAD
SANJANA R	DEVELOPER
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RUHI SHARMA	DEVELOPMENT LEAD
MEGHANA U	TEST LEAD
SAGARIKA DS	CUSTOMER CONTACT LEAD

Modified Problem Statement

Small-scale farmers are unable to prove the origin, quality, and authenticity of their agricultural produce.
Due to lack of transparency in the supply chain, buyers do not fully trust product claims, leading to lower prices and reduced income for farmers.

User/ Customer Understanding

We interacted with small-scale farmers and local buyers to understand real-world challenges.

Key insights:

- Buyers want proof of origin and quality
- Farmers struggle to justify higher prices
- Trust is the biggest gap in the supply chain
- Users prefer simple and easy-to-use systems

Our Solution

AgroLedger is a hybrid agricultural provenance system that enables:

- Farmers to register product details digitally
- Generation of QR codes for each batch
- Buyers to verify product authenticity instantly

The system combines blockchain and cloud technologies to ensure both trust and accessibility.

SYSTEM ARCHITECTURE

AgroLedger – Hybrid Agricultural Provenance System

1. PRESENTATION LAYER

React Frontend (Vite)

- User Interface (Web Browser)
- Register new batch
- Search batch by ID
- Generate QR Code
- Read URL parameters (?batch=FT-123)
- Display batch details

↓ **HTTP Requests**

2. SERVICE LAYER

JSONBin REST API

- Acts as backend service
- Handles data storage & retrieval
- Used Endpoints:
  - GET /v3/b/{BIN_ID}/latest → Fetch data
  - PUT /v3/b/{BIN_ID} → Save data
- Authentication: **X-Master-Key** (API Key)

↓ **HTTP Requests**

3. DATA LAYER

JSON Storage (Database)

- Stores all batch records
- Data is stored as JSON
- Accessible across devices
- Acts as cloud-based ledger

DATA FLOW

Register Batch Flow

User registers batch → Frontend creates batch object
→ Fetch existing data → Append new batch
→ Save updated data to JSONBin → Generate QR

Search Batch Flow

User enters Batch ID → Fetch all batches from JSONBin
→ Search in array → If found, display details
→ If not found, show "Batch not found"

QR Scan Flow

QR Code scanned → Opens URL (/?batch=FT-123)
→ Frontend reads batch ID → Auto-search batch
→ Display result to user

KEY POINTS

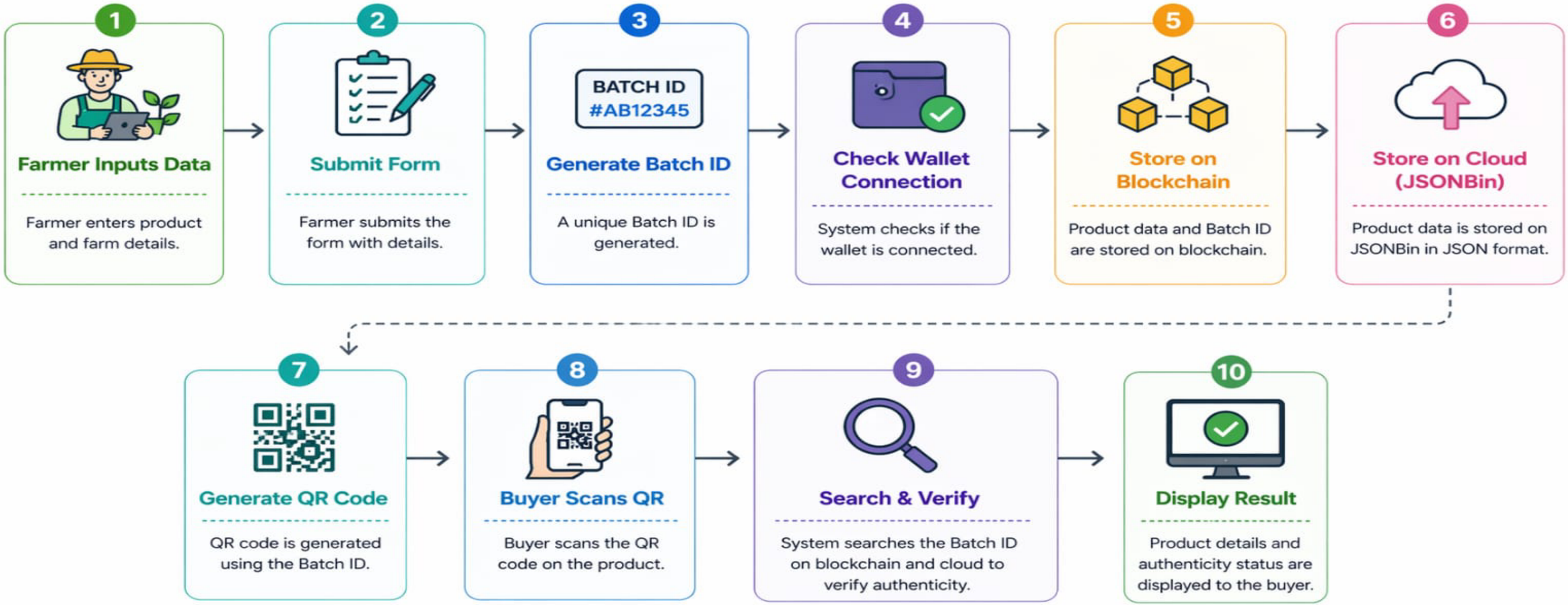
- 3-Tier Architecture: Presentation, Service, Data
- Cloud-based storage ensures cross-device access
- QR-based verification makes the system easy to use
- Simple, scalable and cost-effective solution

SUMMARY

User (Browser) → React Frontend → JSONBin API → JSON Storage

Register / Search → Data stored in cloud → QR generated → Data retrieved → Result displayed

Full Product Flow



Key Features

- Hybrid storage (Blockchain + Cloud)
- QR-based product verification
- No login required for buyers
- Cross-device accessibility
- Simple interface for farmers
- Real-time product tracking

Use Cases & High level Requirements

1. **UC1: Farmer Registers Batch**
   Input: Crop details (name, location, date)
   Output: Unique Batch ID + QR Code generated
2. **UC2: Buyer Verifies Product**
   Input: QR code scan
   Output: Product details displayed (origin, quality, timeline)
3. **UC3: Batch Search & Verification**
   Input: Batch ID or QR link
   Output: Data retrieved from cloud and shown with status
4. **UC4: Cross-Device Access**
   Input: QR shared via link or WhatsApp
   Output: Buyer accesses product data instantly without login

Requirements document available on GitHub — includes Functional Requirements , Non-Functional Requirements (performance, cost, usability), and Acceptance Criteria for each feature.

Proof of Concept (PoC) / Demo

Github: Github

Conclusion

AgroLedger improves transparency in agricultural supply chains by enabling farmers to register produce and generate QR codes for verification.
It allows buyers to easily access product details, building trust and ensuring fair pricing.
The system is simple, low-cost, and accessible, making it suitable for real-world use.